

POWER WITHOUT POSSESSION

How India's information and communications technology ecosystem is built

LAYERS		SOURCE	MAJOR COUNTRIES
	Enterprise Software	Microsoft, Oracle, SAP, Salesforce, etc	• United States • Germany
	Cybersecurity	CrowdStrike, Palo Alto Networks, Cisco	• United States
	Cloud Infrastructure	Microsoft Azure, Amazon Web Services, Google Cloud	• United States
	Servers and Data Centers	Dell, HPE	• United States
	High-Performing Compute	NVIDIA, AMD	• United States
	Electronic Components/ IT Hardware	An estimated 70-80% of key electronic components, including laptops, are sourced from overseas	• China • Hong Kong • Taiwan
	Semiconductors	Nearly 90-95% of semiconductor demand is met through imports	• China • Taiwan • South Korea

56 |

dependence. The world is less globalised today than it was in 2015, making resilience and technological sovereignty far more important than they once seemed," Srikanth Velamakanni, Group Chief Executive, Fractal, and Chairperson, Nasscom, tells *BT*.

GAUGING CAPABILITY

India today generates nearly 20% of the world's data, yet stores only about 3% of it domestically, a stark reminder of the gap between digital consumption and technological self-reliance.

The contrast is equally visible in electronics manufacturing. The country has emerged as the world's second-largest smartphone manu-

facturer and, in 2025, overtook China to become the largest exporter of smartphones to the United States. But how much of the underlying technology is made in India?

From display panels and sensors to semiconductors and advanced electronic components, a significant portion of the supply chain continues to depend on imports, primarily from China and Taiwan. While assembly operations have scaled rapidly, domestic value addition remains limited in several critical segments.

Pankaj Mohindroo, Chairman of the India Cellular & Electronics Association, says semiconductors remain India's biggest vulnerability. "Wherever the CPU is being

imported, that has to be changed quickly. That alone can ensure 15-20% value addition. Beyond the most advanced semiconductor nodes, India will have to develop capabilities across the entire electronics value chain," he says.

India's vast talent pool and engineering capabilities have helped it establish a strong position in software and digital services. However, dependence rises sharply lower down the technology stack.

Ken Research says India relies on overseas vendors —mostly from the US—for about 65% enterprise software requirements and nearly 80% cybersecurity solutions. Dependence is higher in hardware and electronics, with imports ac-